FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – II

JEE (Main)-2020

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 75 questions.
- Each subject (PCM) has 25 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only three sections: **Section-A**, **Section-B and Section-C**.

  Section-A (01 – 20, 26 – 45, 51 – 70) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 22, 46 – 47, 71 – 72) *contains 6 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

  Section-C (23 – 25, 48 – 50, 73 – 75) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

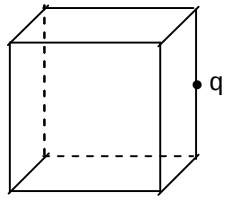
PART – I

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. An object of mass 40 kg and having velocity 4 m/s collides with another object of mass 60 kg, having velocity 2 m/s in the same direction. The loss of energy when the collision is perfectly inelastic is
   (A) 392 J (B) 440 J
   (C) 48 J (D) 110 J

2. The total flux of electric lines of forces of a charge q placed at the mid point of the edge of a side of rectangular box, as shown in figure through the box is

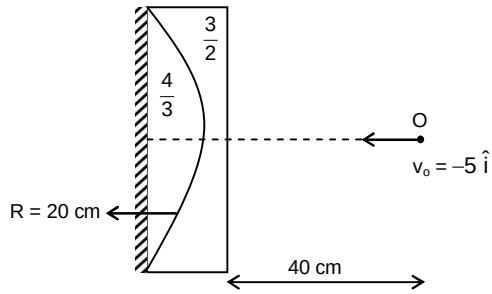


   (A) $\frac{q}{2\in_o}$ (B) $\frac{q}{4\in_o}$
   (C) $\frac{q}{8\in_o}$ (D) $\frac{q}{16\in_o}$

3. 1 gm of a radioactive substance takes. 50 sec to loose 0.01 gm then the half life of the sample will be
   (A) $\frac{50\ln(2)}{\ln\left(\frac{100}{99}\right)}$ (B) $\frac{50\ln(2)}{\ln(100)}$
   (C) $\frac{50\ln(2)}{\ln(99)}$ (D) $\frac{50\ln(2)}{\ln(0.99)}$

Space for Rough work

4. The lens combination having refractive indices $\frac{4}{3}$ & $\frac{3}{2}$ as shown in the figure, is silvered. The velocity of image will be?



(A) $1.8\,\hat{i}$

(B) $3\,\hat{i}$

(C) $3.6\,\hat{i}$

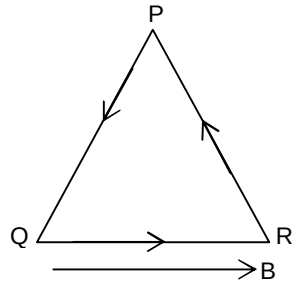
(D) $5\,\hat{i}$

5. Two soap bubbles of radii 2mm and 4mm are brought in contact. If the surface tension of liquid is 7×10^{-2} Nm^{-1}. Then the radius of the common surface is

(A) 2×10^{-3}m (B) 4×10^{-3}m

(C) 6×10^{-3}m (D) 8×10^{-3}m

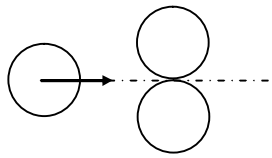
6. An equilateral triangular loop PQR of side l carries a currents i in the direction shown. The loop is kept in uniform magnetic field B, directed parallel to the base of triangle QR as shown. Net torque τ acting on loop is



(A) $F = \frac{\sqrt{3}}{2}\mu^2 B$ (B) $F = \sqrt{3}\ ilB$

(C) $\tau = 0$ (D) $\tau = \frac{\sqrt{3}l^2iB}{4}$

7. Two smooth identical stationary spheres are kept touching each other on a smooth horizontal floor as shown. A third identical sphere moving horizontally with a constant speed hits both stationary spheres symmetrically. If after collision the third sphere moves in same direction with one fourth of its initial speed , the coefficient of restitution will be



(A) 2/3 (B) 1/3

(C) 1/4 (D) 1/6

Space for Rough work

8. A beaker containing a liquid of density ρ moves up with an acceleration a. The pressure due to the liquid at a depth h below the free surface of the liquid is

(A) $h\rho g$ (B) $h\rho(g-a)$

(C) $h\rho(g+a)$ (D) $2h\rho g\left(\frac{g+a}{g-a}\right)$

9. In a radioactive reaction an unstable nucleus A dis-integrates into a stable nucleus B. But A is generated at a constant rate of q nucleus per second. Then at steady state number of nucleus of A will be

$q\ (sec^{-1}) \longrightarrow A \xrightarrow{\lambda} B$

(Parent Nucleus) (Daughter Nucleus)

(A) $q\lambda$ (B) $\frac{q}{\lambda}$

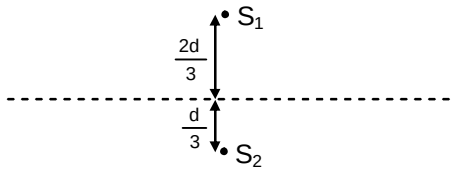
(C) $q-\lambda$ (D) $\frac{\lambda}{q}$

10. A particle moves in x-y plane such that its position vector varies with time as $\vec{r}=(2\sin 3t)\hat{i}+2(1-\cos 3t)\hat{j}$. The equation of trajectory of the particle will be

(A) $x^2+y^2=1$ (B) $x^2+(2-y)^2=4$

(C) $(2-x)^2+y^2=4$ (D) $(1-x)^2+(1-y)^2=4$

11. Two coherent point sources of frequency ($f=\frac{10v}{d}$ where v is speed of sound in air) are placed at a distance d apart as shown in figure. The receiver is free to move along the dotted line shown in the figure. Find total number of maxima heard by receiver.



(A) 6 (B) 7

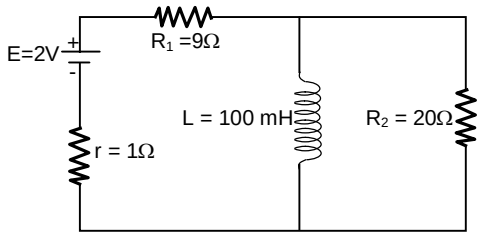
(C) 5 (D) 8

Space for Rough work

12. In the displacement method, magnification for two positions of the lens are 2 and 0.5 and the distance between the two positions of the lens is 30 cm. If the focal length of lens is
(A) 15 cm (B) 20 cm
(C) 25 cm (D) 30 cm

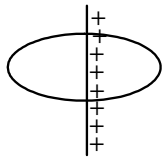
13. A concave mirror of focal length 15 cm is placed in water of refractive index $\mu = 4/3$. Water is filled upto 30 cm. An object of height 1 cm is placed at a distance of 20 cm in front of the mirror. If the size of image is n times size of object, then n =
(A) 3 (B) 2
(C) 2.5 (D) 5

14. The value of time constant for the electric circuit shown in the figure is: (r is the internal resistance of the cell of emf E)
(A) 15s
(B) 1.5s
(C) 0.15s
(D) 0.015s



15. A simple circuit contains an ideal battery and a resistance R. If a second resistor is placed in parallel with the first,
(A) the potential across R will decrease
(B) the current through R will decreased
(C) the current delivered by the battery will increase
(D) the power dissipated by R will increased

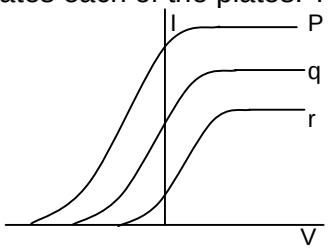
16. A very long uniformly charged rod falls with a constant velocity *V* through the center of a circular loop. Then the magnitude of induced emf in loop is (charge per unit length of rod = λ)

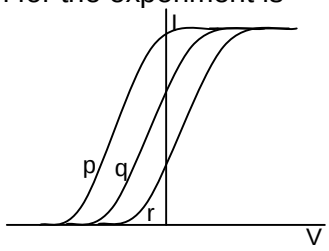


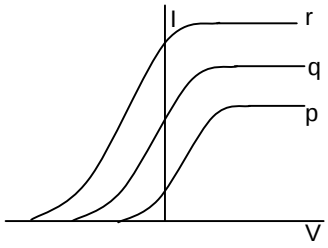
(A) $\frac{\mu_0}{2\pi}\lambda V^2$ (B) $\frac{\mu_0}{2}\lambda V^2$
(C) $\frac{\mu_0}{2\lambda}V$ (D) zero

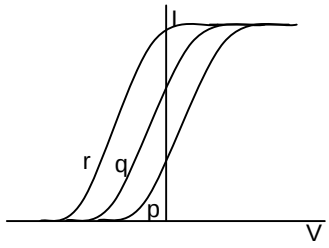
Space for Rough work

17. Photoelectric effect experiments are performed using three different metal plates p, q and r having work functions ϕ_p = 2.0 eV, ϕ_q = 2.5 eV and ϕ_r = 3.0 eV, respectively. A light beam containing wavelengths of 550 nm, 450 nm and 350 nm with equal intensities illuminates each of the plates. The correct I-V graph for the experiment is

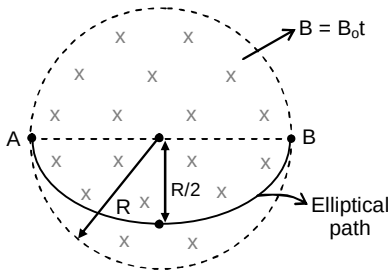
(A) 

(B) 

(C) 

(D) 

18. There is a uniform time varying magnetic field in a circular region as shown in the figure. Find out the potential difference across 2 point along an elliptical path as shown in figure.

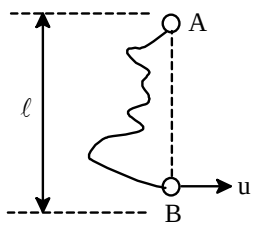


(A) $\frac{\pi R^2}{2}B_o$ (B) $\frac{\pi R^2}{2}B_o$

(C) $\frac{\pi R^2}{4}B_o$ (D) $\frac{\pi R^2}{5}B_o$

Space for Rough work

19. Two balls A & B both of mass m & connected by a light inextensible string of length 2ℓ. Whole system is on a frictionless horizontal table. Ball B is given a velocity u (as shown) $\perp$r to AB. The velocity of ball A just after the string becomes taut is



(A) $\frac{u\sqrt{3}}{4}$ (B) $u\sqrt{3}$

(C) $\frac{u\sqrt{3}}{2}$ (D) $\frac{u}{2}$

20. A block of mass 2 kg rest on a rough inclined plane making an angle 30° with horizontal of coefficient of friction 0.8. (g = 10 m/s^2). The magnitude of resultant of friction and normal reaction on the block will be

(A) $\sqrt{292}$ N (B) $\sqrt{364}$ N

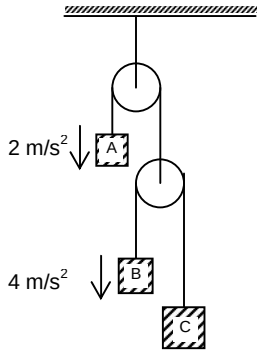
(C) 20 N (D) 18 N

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

21. Water rises in a capillary tube to a height of 2 cm, In another capillary whose radius is one third of it, how high the water will rise (in cm)?

22. In the figure (in m/s^2) shown acceleration of blocks A and B are as shown. Find the magnitude of acceleration of block C.



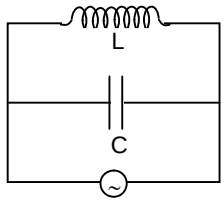
Space for Rough work

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

23. A hollow sphere and a solid sphere have equal mass and equal moment of inertia about the respective diameter. Find the ratio of their square of radii is given by

24. For the circuit shown in the figure, the current through the inductor is 0.6 A, while the current through the capacitor is 0.4 A. Find the current (in A) drawn from the generator



Generator

25. If 10% of the current passes through a moving coil galvanometer of resistance 99 ohm, then the shunt resistance will be:

Space for Rough work

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

26. $KO_2 + S \xrightarrow{\Delta} (P) \xrightarrow{BaCl_2} (Q)\downarrow (\text{White ppt.})$

$(S)\uparrow_{gas} \xleftarrow{HCl} (R) \xleftarrow[\Delta]{\text{Carbon}}$ (Q)

The unknown compounds P, Q, R and S in the above reaction sequence are respectively

(A) K_2SO_3, $BaSO_3$, BaS, H_2S (B) K_2S, BaS, CS_2, H_2S

(C) K_2SO_4, $BaSO_4$, BaS, H_2S (D) K_2SO_4, $BaSO_4$, $BaCO_3$, CO_2

27. Find ΔH at 358 K for the reaction,

$Fe_2O_3(s) + 3H_2(g) \longrightarrow 2Fe(s) + 3H_2O(\ell)$

Given that, $\Delta H_{298} = -33.29$ kJ/mole and C_p for $Fe_2O_3(s)$, Fe(s), $H_2O(\ell)$ and $H_2(g)$ are 103.8, 25.1, 75.3 and 28.8 J/K mole.

(A) 85.9 kJ/mole (B) –28.136 kJ/mole

(C) +28.136 kJ/mole (D) 85.9 J/mole

28. An acid-base indicator has $K_a = 3 \times 10^{-5}$. The acid form of the indicator is red and the basic form of the indicator is blue. By how much must the pH change in order to change the indicator from 75% red to 75% blue? [log 3 = 0.4770]

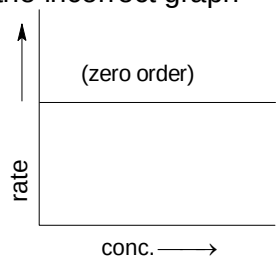
(A) 4.05 (B) 0.95

(C) 5 (D) 9

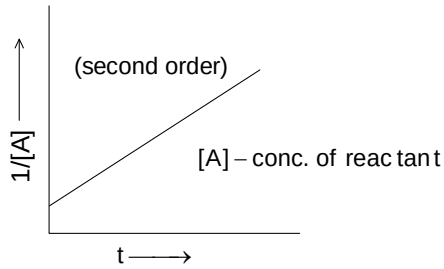
Space for Rough work

29. Select the incorrect graph

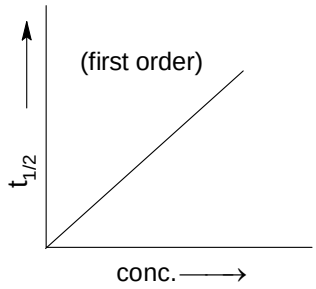
(A)



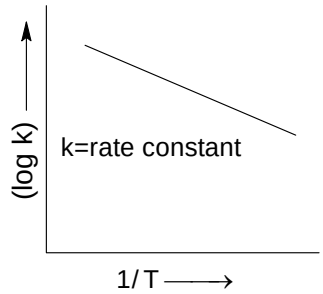
(B)



(C)



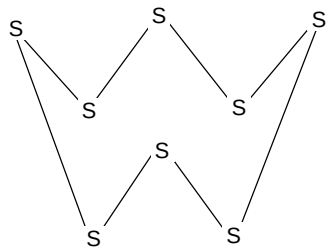
(D)



30. For the reaction,
$C_7H_8(l) + 9O_2(g) \rightarrow 7CO_2(g) + 4H_2O(l)$, calculated heat of reaction is 232.6 kJ/mol and observed heat of reaction is 50.4 kJ/mol, then the resonance energy is
(A) – 182.2 kJ / mol (B) + 182.2 kJ / mol
(C) 172 kJ/ mol (D) None of these

31. Heat of neutralization of four monobasic acids A, B, C and D are -13.7, - 9.4, - 11.2 and – 12.4 Kcal / mole respectively, when they are neutralized by a common base. The acidic character obeys the order
(A) A > B > C > D (B) A > D > C > B
(C) D > C > B > A (D) D > B > C > A

32. The oxidation number and hybridization of sulphur in S_8 molecule are respectively:
(A) +2 and sp^2
(B) +2 and sp^3
(C) zero and sp^2
(D) zero and sp^3



Space for Rough work

33. The relation between the molar conductance ($\wedge_m$) and equivalent conductance ($\wedge_e$) of the solution of $MgCl_2.KCl.6H_2O$ is

(A) $\wedge_m = \wedge_e \times 8$ (B) $\wedge_m = \wedge_e \times 3$

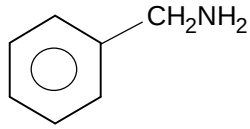
(C) $\wedge_e = \wedge_m \times 3$ (D) $\wedge_e = \wedge_m \times 8$

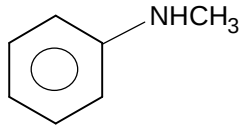
34. For which of the following aqueous solution, there is maximum difference between the normal and abnormal colligative properties?

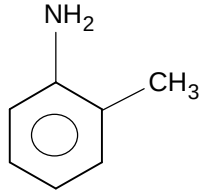
(A) 0.1 m $C_6H_{12}O_6$ (B) 0.1 m CH_3COOH

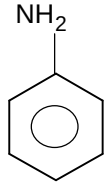
(C) 0.1 m HCOOH (D) 0.1 m CH_3OH

35. Which of the following is the strongest base?

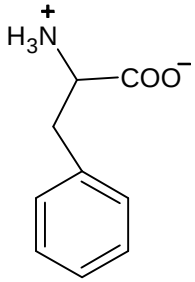
(A) 

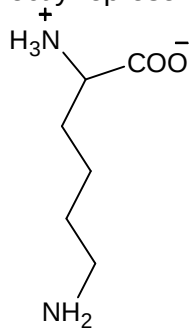
(B) 

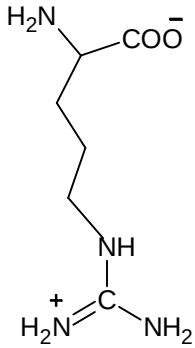
(C) 

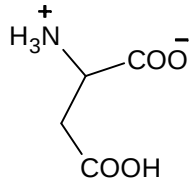
(D) 

36. Which of the following zwitter ionic form is not correctly represented for an amino acid?

(A) 

(B) 

(C) 

(D) 

Space for Rough work

37. How many effective no. of Na^+ ions are present in the unit cell of NaCl if the ions along all the body diagonals of the unit cell are removed?
(A) 4 (B) 3
(C) 2 (D) 1

38. Which one of the following has aromatic character?
(A) Cyclopentadienyl cation (B) Cyclopentadienyl radical
(C) Cyclopentadienyl anion (D) Cyclopentadiene

39. $Zn(s)|\ Zn^{2+}(1M)||\ Cu^{2+}(1M)|Cu(s)$
$E^0_{Zn^{2+}/Zn} = -0.76\,V$ and $E^0_{Cu^{2+}/Cu} = 0.34\,V$
Which of the following relation(s) is/are correct for the above electrochemical cell?
(A) $E^0_{cell} > 0$ (B) $E_{cell} > 0$
(C) $E_{cell} = E^0_{cell}$ (D) All are correct

40. Number of C = C bonds present in anthracene is:–
(A) 5 (B) 3
(C) 9 (D) 7

41. Which of the following interhalogen compound does NOT exist?
(A) IF_3 (B) IF_4
(C) IF_5 (D) IF_7

42. The fraction of chlorine precipitated by $AgNO_3$ solution from $[Cr(NH_3)_5Cl]Cl_2$ is:–
(A) $\frac{1}{2}$ (B) $\frac{1}{3}$
(C) $\frac{2}{3}$ (D) $\frac{1}{4}$

Space for Rough work

43. The following two equilibria exist simultaneously in a closed container of volume 2 L

$A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$

$A_2(g) + C(g) \rightleftharpoons A_2C(g)$

Which of the following change does not affect the concentration of AB(g)

(A) Increase in pressure
(B) Addition of C
(C) Addition of He gas at constant volume
(D) All of these

44. In borax bead test, which compound is formed?

(A) Meta borate (B) Ortho borate
(C) Tetra borate (D) Double oxide

45. The normal boiling point of a liquid 'X' is 500 K. Which of the following statement(s) is/are incorrect for the following process?

$X(\text{liquid}) \longrightarrow X(\text{gas})$

(A) At 500 K and 1 atm pressure, $\Delta G = 0$
(B) At 500 K and 1.5 atm pressure, ΔG = +ve
(C) At 500 K and 0.2 atm pressure, ΔG = -ve
(D) At 520 K and 1 atm pressure, ΔG = +ve

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

46. How many visible lines are emitted during transition from 5th orbit to ground state in hydroen emission spectrum?

47. For the reactoin at $25^{o-}C$

$A(g) + 2B(g) \rightleftharpoons 2C(g) + D(g)$

The intial concentratoin of B is 1.5 times the metal conc. of A. If at equilirbium, the concentration of A and D are equal. The equilibrium constant at 25^oC is

Space for Rough work

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

48. Find the wavelength of light (in $\overset{o}{A}$) emitted by an electron during transition from n = 3 to n = 1 level in C^{+5} ion. Given $R_H = 1.077 \times 10^7\ m^{-1}$

49. The sum of lone pairs on central atom of SF_4, SF_6 and H_2O is X. Calculate X/2.

50. In Na^+, Mg^{2+}, Al^{3+}, O^{2-}, F^-, Cl^- atoms. If x atoms have more radius as compared to Al^{3+} then find $\frac{x}{2}$?

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

51. A function y = f(x) satisfies $(x+1).f'(x)-2(x^2+x)f(x)=\frac{e^{x^2}}{(x+1)}$, $\forall x > -1$. If f(0) = 5, then f(x) is

(A) $\left(\frac{3x+5}{x+1}\right).e^{x^2}$ (B) $\left(\frac{6x+5}{x+1}\right).e^{x^2}$

(C) $\left(\frac{6x+5}{(x+1)^2}\right).e^{x^2}$ (D) $\left(\frac{5-6x}{x+1}\right).e^{x^2}$

52. The relation R on the set of complex numbers, defined by z_1Rz_2 if and only if $\frac{z_1-z_2}{z_1+z_2}$ is real, is

(A) reflexive but not symmetric (B) reflexive but not transitive

(C) symmetric but not transitive (D) an equivalence relation

53. If $\alpha \in \left(0, \frac{\pi}{2}\right)$, then the expression $\sqrt{x^2+x}+\frac{\tan^2 x}{\sqrt{x^2+x}}$ is always greater than or equal to

(A) $2\tan\alpha$ (B) 2

(C) 1 (D) $\sec^2\alpha$

54. Let the function $f(x)=x^2+x+\sin x-\cos x+\log(1+|x|)$ be defined on the interval $[0, 1]$. The function $g(x)$ on $[-1, 1]$ satisfying $g(-x)=-f(x)$ is

(A) $x^2+x+\sin x+\cos x-\log(1+|x|)$ (B) $-x^2+x+\sin x+\cos x-\log(1+|x|)$

(C) $-x^2+x+\sin x-\cos x+\log(1+|x|)$ (D) none of these

Space for Rough work

55. Let $a = a_1i + a_2j + a_3k$, $b = b_1i + b_2j + b_3k$ and $c = c_1i + c_2j + c_3k$ be three non zero vectors such that c is a unit vector perpendicular to both a and b. If the angle between a and b is $\frac{\pi}{6}$, then $\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}^2$ is equal to

(A) 0

(B) 1

(C) $\frac{1}{4}\left(a_1^2 + a_2^2 + a_3^2\right)\left(b_1^2 + b_2^2 + b_3^2\right)$

(D) $\frac{3}{4}\left(a_1^2 + a_2^2 + a_3^2\right)\left(b_1^2 + b_2^2 + b_3^2\right)\left(c_1^2 + c_2^2 + c_3^2\right)$

56. If $I = \int_0^{\pi} e^{\left|\left(\frac{1}{2}\right)\cos x\right|}\left\{2\sin\left(\frac{1}{2}\cos x\right) + 3\cos\left(\frac{1}{2}\cos x\right)\right\}\sin x\,dx$, then I equals

(A) $7\sqrt{e}\cos\left(\frac{1}{2}\right)$ (B) $7\sqrt{e}\left[\cos\left(\frac{1}{2}\right) - \sin\left(\frac{1}{2}\right)\right]$

(C) 0 (D) none of these

57. If x is real, then the maximum value of $y = 2(a - x)\left(x + \sqrt{x^2 + b^2}\right)$

(A) $a^2 + b^2$ (B) a^2

(C) $\sqrt{a^2 + b^2}$ (D) $a\sqrt{a^2 + b^2}$

58. If $\tan^{-1}\frac{1}{1+2} + \tan^{-1}\frac{1}{1+(2)(3)} + \tan^{-1}\frac{1}{1+(3)(4)} + \ldots\ldots + \tan^{-1}\frac{1}{1+n(n+1)} = \tan^{-1}\theta$, then $\theta =$

(A) $\frac{n}{n+1}$ (B) $\frac{n+1}{n+2}$

(C) $\frac{n}{n+2}$ (D) $\frac{n-1}{n+2}$

Space for Rough work

59. If $x_{n+1} = \sqrt{\frac{1}{2}(1+x_n)}$, then $\cos\left[\frac{\sqrt{1-x_0^2}}{x_1x_2x_3.....\text{to infinite}}\right] (-1< x_0 <1)$ is equal to

(A) –1 (B) 1

(C) x_0 (D) $\frac{1}{x_0}$

60. If $f(x) = x^2 + 2bx + 2c^2$ and $g(x) = -x^2 - 2cx + b^2$ are such that min. $f(x) >$ max g(x), then the relation between b and c is

(A) no relation (B) $0 < c < \frac{b}{2}$

(C) $|c| < |b|\sqrt{2}$ (D) $|c| > |b|\sqrt{2}$

61. Two unit vectors $\vec{a}$ and $\vec{b}$ having angle θ between them. Now $\phi(\theta) = \int\limits_{-(\vec{a}.\vec{b})^2}^{|\vec{a}\times\vec{b}|^2} f^2(x).dx$

and f satisfies the condition $f(x) + f(y) = \frac{x+y}{xy} \forall x, y \in R - \{0\}$. Then the fundamental period of $\phi(\theta)$ is

(A) Not Defined (B) π

(C) 2π (D) $\frac{\pi}{2}$

62. Consider parabola $y^2 = 40x$, A circle $S = 0$ is drawn taking one of its focal chord as diameter. The extremeties of the focal chord being P_1 and P_2

(A) A point 'P' is taken on the circumference of S = 0, Then $\angle P_1PP_2 = \frac{\pi}{4}$

(B) x = 10 is a tangent to S = 0

(C) Combined equation of the tangents at P_1 and P_2 has (coefficient x^2) + (coefficient y^2) = –2

(D) ΔP_1PP_2 has its orthocentre which lies on it

Space for Rough work

63. The remainder when $x = 5^{5^5}$ is divided by 24 is

(A) 1 (B) 3
(C) 5 (D) 23

64. Let f be a function with domain $[-5, 5]$ and let $g(x) = |3x + 4|$. Then the domain of $(fog)(x)$ is

(A) $\left(-3, \frac{1}{3}\right)$ (B) $\left[-3, \frac{1}{3}\right]$
(C) $\left(-3, \frac{1}{3}\right]$ (D) N.O.T.

65. Sum of maximum and minimum values of $\left(\sin^{-1} x\right)^4 + \left(\cos^{-1} x\right)^4$ is _____

(A) $\frac{137\pi^2}{128}$ (B) $\frac{\pi^4}{17}$
(C) $\frac{17\pi^4}{16}$ (D) $\frac{137\pi^4}{128}$

66. Number of solutions of the equation $3\tan x + x^3 = 2$ in $\left(0, \frac{\pi}{4}\right)$, are

(A) 0 (B) 2
(C) 3 (D) None of these

67. If the vectors $\vec{b} = \left(\tan\alpha, -1, 2\sqrt{\tan\frac{\alpha}{2}}\right)$, $\vec{c} = \left(\tan\alpha, \tan\alpha, \frac{-3}{\sqrt{\tan\frac{\alpha}{2}}}\right)$ are orthogonal and vectors $\vec{a} = (1, 3, \sin 2\alpha)$ makes obtuse angle with the z axis the α equals

(A) $(2n+1)\pi + \tan^{-1} 2$ (B) $(2n+1)\pi - \tan^{-1} 2$
(C) $n\pi - \tan^{-1} 2$ (D) None of these

Space for Rough work

68. The Real value of 'm' for which the substitution $y = u^m$ will transform the differential equation $2x^4y\frac{dy}{dx} + y^4 = 4x^6$ into a homogeneous equation, is

(A) $\frac{1}{2}$ (B) 1

(C) $\frac{3}{2}$ (D) 2

69. Sign of $(x-1)(x-3)(x-4)(x-6)+10$ is

(A) Positive for all real values of x (B) Negative for all $x \in R$

(C) Can be positive, negative or zero (D) Can't say

70. A parallelogram is constructed with its sides parallel to the asymptotes of the hyperbola $x^2 - 2y^2 - 4x + 4y - 2 = 0$ and one of its diagonals is a chord of the same hyperbola. If the other diagonal passes through a fixed point (a, b) $a,b \in R$ then

(A) $a + b = 4$ (B) $a - b = 3$

(C) $\sum_{r=1}^{a+b} r = 6$ (D) $\sum_{r=1}^{a-b} r = 6$

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

71. An urn contain 3 red balls and n white balls. Mr. A draws two balls together from the urn. The probability that they have the same colour is 1/2. Mr. B draws one ball from the urn, notes its colour and replaces it. He then draws a second ball from the urn and finds probability that both balls have the same colour is $\frac{5}{8}$. The possible value of n is_____

72. The 20th term of an arithmetic sequence is $\log_{10} 20$ and the 32nd term is $\log_{10} 32$. If exactly one term of the arithmetic sequence is a rational number which in lowest form is $\frac{p}{q}; (p, q \in N)$ then (p + q) = _____

Space for Rough work

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

73. If $f:[0,1]\to R$ is a continuous function satisfying $\int_0^1 f(x)dx=\frac{1}{3}+\int_0^1\left(f\left(x^2\right)\right)^2 dx$, then the reciprocal of $\frac{1}{8}f\left(\frac{1}{4}\right)$ is _____

74. Any chord of conic $x^2+y^2+xy=1$ passing through origin is bisected at point (p, q), then $\frac{(p+q+1)}{8}$ equals _____

75. Let z and ω be complex numbers such that $z+\omega=i$ and $z^2+\omega^2=1$. If area of triangle formed by z, ω and origin is equal to A, then find the value of $10A^2$.

Space for Rough work